GenoDisPred: Computational Prediction of Protein Disorder & Genome-wide prediction of intrinsic disorder in E. Coli.

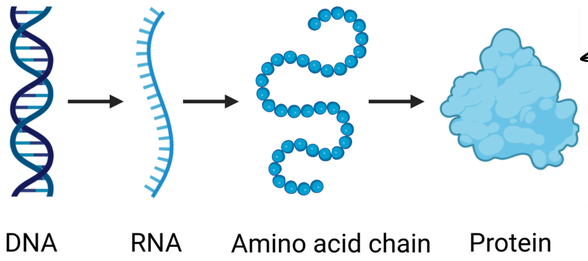
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Introduction

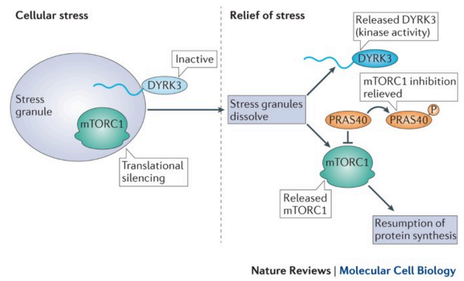


ANFINSEN's DOGMA

Polypeptide chain folds into a unique native structure

Biological significance of Disordered Proteins

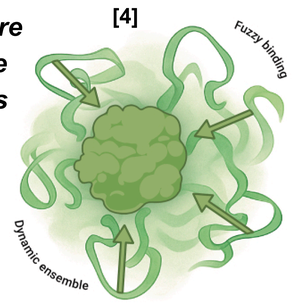
[3]



Dynamic signalling & regulation

Flexible structure allows multiple conformations

[4]

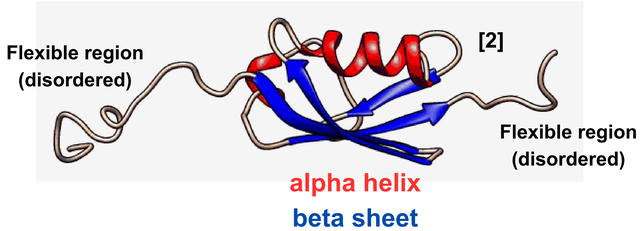


Research Gap

There are no available tools that can

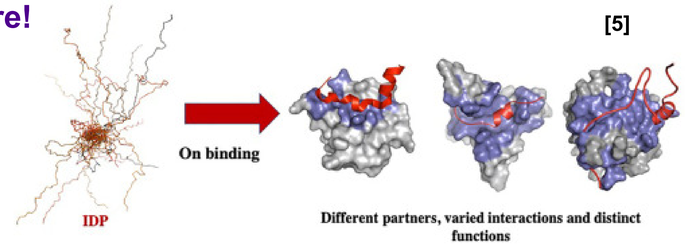
- **Provide disorder information directly from nucleotide sequence.**
- **Perform large-scale genome-wide prediction without sacrificing accuracy or computational efficiency.**

Not all proteins conform to a rigid 3D structure!



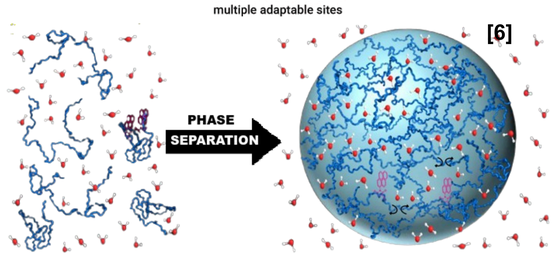
▲ **Conformational flexibility in SUMO-1 protein (PDB 1A5R)**

[5]



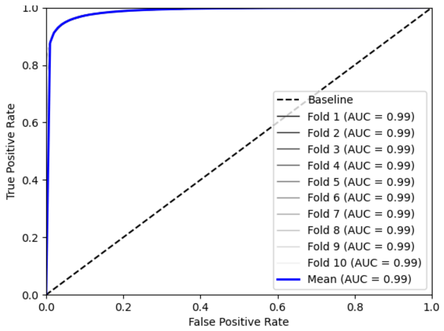
High promiscuity & High Specificity

[6]



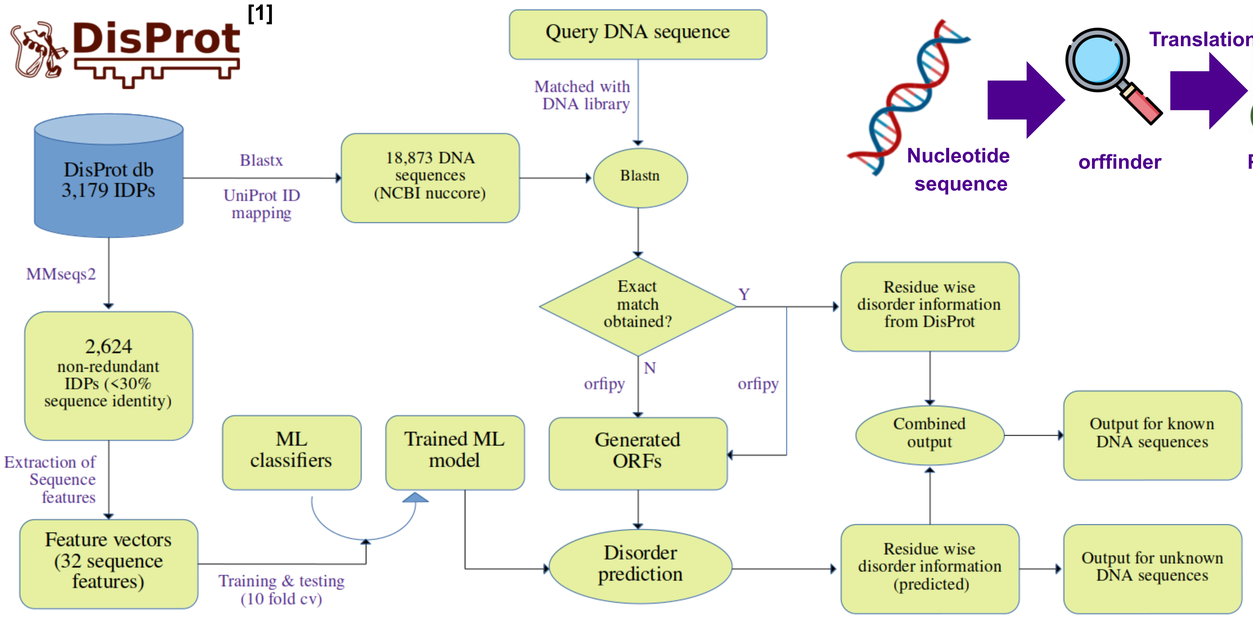
Liquid Liquid Phase Separation

Results

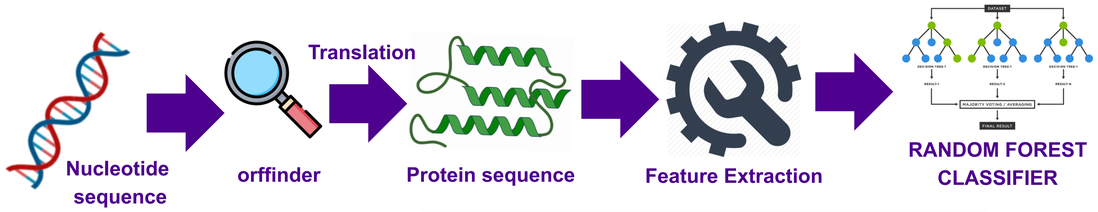


ROC curve across 10 cross validations

Prediction Pipeline



Model Architecture



Sliding Window Features

Physiochemical	Composition	Secondary-structure
• Avg. mol. wt. • Avg. pI • Mean electronegativity • Sequence variability	• Std. Aā % • Acidic residues % • Polar residues %	• Chou Fasman secondary structure probabilities

Final Feature Vector = 32 dim.

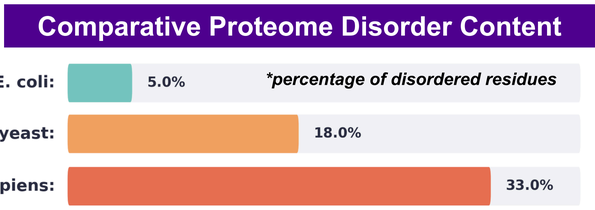
Performance

Accuracy	**= 0.948**	**F1 score**	**= 0.833**
Precision	**= 0.991**	**MCC**	**= 0.818**
Recall	**= 0.719**	**ROC-AUC**	**= 0.985**

E. coli. Genome Analysis



Comparative Proteome Disorder Content



References

[1] DisProt: https://doi.org/10.1093/nar/gkaf1175
[2] SUMO-1 NMR structure: rcsb.org/structure/1A5R
[3] Signalling & regulation: doi.org/10.1038/nrm3920
[4] Modes of binding: doi.org/10.1098/rsob.210222
[5] Binding: doi.org/10.1016/j.bpc.2022.106769
[6] LLPS: doi.org/10.1021/acs.jpclett.9b01731

Acknowledgements

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